

Aptitude Assessment 1 (AT1)

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OFFICIAL ANSWER KEY & DETAILED SOLUTIONS

Test Code: AT1 • Total Questions: 150 • Max Marks: 150

Duration: 90 Minutes

Format: Multiple Choice Questions

Question 1

[1 Mark]

Three numbers are in the ratio of 2 : 3 : 5 and their L.C.M. is 1800. Their H.C.F. is:

A 30

B 60

✓ Correct Answer

C 90

D 120

Explanation: Let the numbers be $2x$, $3x$, $5x$. Their LCM is $30x = 1800 \Rightarrow x = 60$. The HCF is $x = 60$.

Question 2

[1 Mark]

The ratio of two numbers is 4 : 5 and their H.C.F. is 6. Their L.C.M. is:

A 60

B 90

C 120

✓ Correct Answer

D 150

Explanation: Numbers are $4 * 6 = 24$ and $5 * 6 = 30$. $\text{LCM}(24, 30) = 120$.

Question 3

[1 Mark]

The L.C.M. of two numbers is 72. The numbers are in the ratio 3 : 4. Then the sum of the numbers is:

A 36

B 42

✓ Correct Answer

C 48

D 54

Explanation: Let the numbers be $3x$ and $4x$. LCM is $12x = 72 \Rightarrow x = 6$. Numbers are 18 and 24. Sum = $18 + 24 = 42$.

Question 4

[1 Mark]

If one-fourth of one-fifth of a number is 16, then three-eighth of that number is:

A 90

B 105

C 120

✓ Correct Answer

D 140

Explanation: $(1/20) * N = 16 \Rightarrow N = 320$. Then $(3/8) * 320 = 120$.

Question 5

[1 Mark]

Three times the first of three consecutive odd integers is 7 more than twice the third. The third integer is:

A 15**B** 17**C** 19

✓ Correct Answer

D 21

Explanation: Let integers be $x, x+2, x+4$. $3x = 2(x+4) + 7 \Rightarrow 3x = 2x + 15 \Rightarrow x = 15$. Third integer = $15 + 4 = 19$.

Question 6

[1 Mark]

The difference between a two-digit number and the number obtained by interchanging the positions of its digits is 45. What is the difference between the two digits of that number?

A 4**B** 5

✓ Correct Answer

C 6**D** Cannot be determined

Explanation: Difference is $(10x + y) - (10y + x) = 9(x - y) = 45 \Rightarrow x - y = 5$.

Question 7

[1 Mark]

A two-digit number is such that the product of its digits is 12. When 36 is added to the number, the digits are reversed. The number is:

A 26

✓ Correct Answer

B 34**C** 43**D** 62

Explanation: Let number be $10x + y$. $xy = 12$. $10x + y + 36 = 10y + x \Rightarrow 9(y - x) = 36 \Rightarrow y - x = 4$. With $xy = 12$, $x = 2$ and $y = 6$. Number = 26.

Question 8

[1 Mark]

The sum of the digits of a two-digit number is 13 and the difference between the digits is 5. What is the two-digit number?

A 49**B** 94**C** 58**D** Cannot be determined

✓ Correct Answer

Explanation: The number could be 49 or 94, since it is not specified whether tens digit or units digit is greater. Hence, cannot be determined uniquely.

Question 9

[1 Mark]

Which one of the following numbers is exactly divisible by 11?

A 28182

✓ Correct Answer

B 28185**C** 315624**D** 415623

Explanation: For 28182: Sum of odd digits = $2 + 1 + 2 = 5$. Sum of even digits = $8 + 8 = 16$. Difference = $5 - 16 = -11$, divisible by 11.

Question 10

[1 Mark]

If the number $532 * 714$ is completely divisible by 9, then the smallest whole number in place of * will be:

A 3**B** 4**C** 5

✓ Correct Answer

D 6

Explanation: Sum of digits = $5 + 3 + 2 + 7 + 1 + 4 = 22$. Nearest multiple of 9 is 27 $\Rightarrow * = 27 - 22 = 5$.

Question 11

[1 Mark]

The value of a printing machine depreciates at the rate of 10% every year. If its present value is Rs. 14,580, then what was the price of the machine three years ago?

A Rs. 18,000**B** Rs. 20,000

✓ Correct Answer

C Rs. 22,000**D** Rs. 25,000

Explanation: $P * (1 - 0.10)^3 = 14580 \Rightarrow P * 0.729 = 14580 \Rightarrow P = 20000$.

Question 12

[1 Mark]

A student erroneously multiplied a number by $\frac{3}{4}$ instead of $\frac{4}{3}$. What is the percentage error in the calculation?

A 36.5%**B** 43.75%

✓ Correct Answer

C 48.2%**D** 56.25%

Explanation: Error = $(\frac{4}{3} - \frac{3}{4}) / (\frac{4}{3}) * 100 = (\frac{7}{12}) * (\frac{3}{4}) * 100 = (\frac{7}{16}) * 100 = 43.75\%$.

Question 13

[1 Mark]

Find the value of 20% of 15% of Rs. 1,200.

A Rs. 28**B** Rs. 32**C** Rs. 36

✓ Correct Answer

D Rs. 40

Explanation: $0.20 * 0.15 * 1200 = 36$.

Question 14

[1 Mark]

The population of a town is decreasing at a rate of 15% per annum. If the population two years ago was 40,000, what is the present population?

A 27,500**B** 28,900

✓ Correct Answer

C 30,200**D** 32,400

Explanation: Present population = $40000 * (0.85)^2 = 40000 * 0.7225 = 28900$.

Question 15

[1 Mark]

Suresh spends 65% of his salary and deposits 20% of his salary in the bank. If he is left with Rs. 3,000, what is his monthly salary?

A Rs. 18,000**B** Rs. 20,000

✓ Correct Answer

C Rs. 22,500**D** Rs. 25,000

Explanation: Remaining = $100\% - (65\% + 20\%) = 15\%$. $15\% \text{ of Salary} = 3000 \Rightarrow \text{Salary} = 3000 / 0.15 = \text{Rs. } 20,000$.

Question 16

[1 Mark]

A fruit seller had some oranges. He sells 45% oranges and still has 495 oranges. Originally, he had:

A 800 oranges**B** 850 oranges**C** 900 oranges

✓ Correct Answer

D 950 oranges

Explanation: $55\% \text{ of Total} = 495 \Rightarrow \text{Total} = 495 / 0.55 = 900 \text{ oranges}$.

Question 17

[1 Mark]

Three candidates contested an election and received 1,200, 4,800, and 6,000 votes respectively. What percentage of the total votes did the winning candidate get?

A 45%**B** 48%**C** 50%

✓ Correct Answer

D 55%

Explanation: Total votes = $1200 + 4800 + 6000 = 12,000$. Winning % = $(6000 / 12000) * 100 = 50\%$.

Question 18

[1 Mark]

The value of a CNC lathe machine depreciates at the rate of 10% per annum. If the cost of the machine at present is Rs. 200,000, then what will be its worth after 2 years?

A Rs. 158,000**B** Rs. 162,000

✓ Correct Answer

C Rs. 166,000**D** Rs. 172,000

Explanation: Worth after 2 years = $200000 * (0.90)^2 = \text{Rs. } 162,000$.

Question 19

[1 Mark]

Passing marks in an examination are 40%. If a candidate gets 85 marks and fails by 15 marks, the maximum marks in the examination are:

A 200**B** 225**C** 250

✓ Correct Answer

D 300

Explanation: Pass marks = $85 + 15 = 100$. 40% of Total = 100 $\Rightarrow$ Total = $100 / 0.40 = 250$.

Question 20

[1 Mark]

The price of an article was first decreased by 25% and then increased by 20%. What was the overall percent decrease or increase?

A 5% increase**B** 10% decrease

✓ Correct Answer

C 8% decrease**D** No change

Explanation: Factor = $(1 - 0.25) * (1 + 0.20) = 0.75 * 1.20 = 0.90 \Rightarrow 10\% \text{ decrease}$.

Question 21

[1 Mark]

A train travelled a certain distance at a uniform speed. Had the speed been 6 km/h more, it would have needed 4 hours less. Had the speed been 6 km/h less, it would have needed 6 hours more. The distance travelled by the train is:

A 720 km

✓ Correct Answer

B 540 km**C** 640 km**D** 840 km

Explanation: Using distance formulas $D = s \cdot (s+6) \cdot 4/6 = s \cdot (s-6) \cdot 6/6 \Rightarrow 4(s+6) = 6(s-6) \Rightarrow 4s + 24 = 6s - 36 \Rightarrow 2s = 60 \Rightarrow s = 30$ km/h. Distance $D = 30 \cdot 36 \cdot 4 / 6 = 720$ km.

Question 22

[1 Mark]

A boat's speed in still water is 20 km/h, while the river flows with a speed of 5 km/h. If the time taken to cover a certain distance upstream is 4 hours more than downstream, find the distance.

A 120 km**B** 150 km

✓ Correct Answer

C 180 km**D** 200 km

Explanation: $D / (u - v) - D / (u + v) = \Delta t$. Solving for D yields the distance.

Question 23

[1 Mark]

The speed of a boat in still water is thrice the speed of the stream. If the boat takes 18.5 sec to go to a certain place downstream, find the additional time required to cover the same distance travelling upstream:

A 18.5 sec

✓ Correct Answer

B 22.5 sec**C** 37.0 sec**D** 15.0 sec

Explanation: Downstream speed = $4v$, upstream speed = $2v$. Time upstream = $2 \cdot$ Downstream time. Hence, additional time = downstream time.

Question 24

[1 Mark]

A boat goes 24 km upstream and 36 km downstream in 6 hours. In 8 hours, it goes 32 km upstream and 48 km downstream. Determine the speed of the boat in still water.

A 10 km/h

✓ Correct Answer

B 8 km/h**C** 12 km/h**D** 14 km/h

Explanation: Upstream speed = $u - v = 8$ km/h, downstream speed = $u + v = 12$ km/h. Boat speed in still water $u = (12 + 8)/2 = 10$ km/h and stream $v = 2$ km/h.

Question 25

[1 Mark]

In a metropolitan area, last year's tourism revenue accounted for 25% of overall revenue, with the remaining \$750,000 coming from tech manufacturing. This year, tourism revenue fell by 60%, while manufacturing rose by 150%. What percentage of this year's manufacturing revenue is the revenue from tourism?

A 4%**B** 5.33%

✓ Correct Answer

C 6.67%**D** 8%

Explanation: Calculate previous revenues, apply respective percentage changes, and compute the ratio.

Question 26

[1 Mark]

In a school, 20% of students are below 8 years of age. The number of students above 8 years is $\frac{2}{3}$ of the students of 8 years of age, which is 60. What is the total number of students in the school?

A 125

✓ Correct Answer

B 150**C** 175**D** 200

Explanation: Students of 8 yrs = N. Students above 8 = $(\frac{2}{3})N$. Total of age ≥ 8 is 80% of student population. Total = $(N + (\frac{2}{3})N) / 0.80$.

Question 27

[1 Mark]

In an election between two candidates, one got 60% of the total valid votes, and 20% of the total votes were invalid. If the total number of votes was 8,000, the number of valid votes that the other candidate received was:

A 2,560

✓ Correct Answer

B 3,200**C** 3,840**D** 4,200

Explanation: Valid votes = 80% of total. Other candidate gets $(100 - \text{winning})\%$ of valid votes.

Question 28

[1 Mark]

The incomes of Alex, Bob, and Charlie are in the ratio 7 : 9 : 12, while their expenses are in the ratio 8 : 9 : 15. If their respective savings are Rs. 200, Rs. 450, and Rs. 150, what is Alex's income?

A Rs. 1,400

✓ Correct Answer

B Rs. 1,200**C** Rs. 1,600**D** Rs. 1,800

Explanation: $7x - 8y = 200$ and $9x - 9y = 450 \Rightarrow x - y = 50$. Substituting yields $x = 200$, $y = 150$. Alex's income = $7 * 200 =$ Rs. 1,400.

Question 29

[1 Mark]

A can complete a task in 12 days and B in 18 days. If they work together for 4 days, what fraction of the work remains unfinished?

A 4/9

✓ Correct Answer

B 5/9**C** 1/3**D** 2/9

Explanation: Work done in 4 days = $4 * (1/d1 + 1/d2)$. Work remaining = $1 - \text{Work done}$.

Question 30

[1 Mark]

A, B, and C can complete a job in 20, 30, and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

A 15 days

✓ Correct Answer

B 12 days**C** 18 days**D** 20 days

Explanation: Calculate 3-day work cycle: Day 1 (A) + Day 2 (A) + Day 3 (A+B+C). Total cycles to complete 1 unit of work.

Question 31

[1 Mark]

Evaluate: $25\% \text{ of } 40 + 25\% \text{ of } 60 - 15\% \text{ of } 200 = ?$

A -5

✓ Correct Answer

B 0**C** -10**D** 5

Explanation: Calculate each percentage term and combine using arithmetic order.

Question 32

[1 Mark]

The population of a city increased from 2,00,000 to 3,00,000 in a decade. The average percentage increase of population per year is:

A 5%

✓ Correct Answer

B 6%**C** 7.5%**D** 8%

Explanation: Total decade increase = 50%. Average per year = $50\% / 10 = 5\%$.

Question 33

[1 Mark]

If A:B is 1:2, B:C is 3:4, and C:D is 2:3, find A:B:C:D.

A 3:6:8:12

✓ Correct Answer

B 2:4:6:9**C** 1:2:3:4**D** 3:5:7:9

Explanation: Multiply ratio terms sequentially to harmonize intermediate values.

Question 34

[1 Mark]

The ratio of two numbers is 7 : 9. If the sum of the numbers is 256, find the difference between the numbers.

A 32

✓ Correct Answer

B 28**C** 36**D** 40

Explanation: Sum parts = a+b. 1 part = Sum / (a+b). Difference = |a - b| * (1 part).

Question 35

[1 Mark]

The ages of three siblings are in the ratio 4 : 7 : 9. If the sum of their ages is 40 years, find the difference between the eldest and the youngest in months.

A 120 months

✓ Correct Answer

B 108 months**C** 96 months**D** 132 months

Explanation: 1 part = Total / SumOfParts. Difference in years = (Eldest - Youngest) * part. Convert to months (* 12).

Question 36

[1 Mark]

Two numbers are respectively 30% and 60% more than a third number. The ratio of the two numbers is:

A 13 : 16

✓ Correct Answer

B 3 : 4**C** 11 : 14**D** 15 : 19

Explanation: Let third be 100. Numbers are 100+p1 and 100+p2. Simplify ratio.

Question 37

[1 Mark]

A sum of money is divided among A, B, C, and D in the ratio 4 : 3 : 6 : 2. If C receives Rs. 1,200 more than D, what is B's share?

A Rs. 900

✓ Correct Answer

B Rs. 750**C** Rs. 1,050**D** Rs. 1,200

Explanation: Difference between C and D is (c - d) parts = Rs. Diff => 1 part = Diff / (c - d). B's share = b * 1 part.

Question 38

[1 Mark]

If $0.6 : x :: 3 : 5$, then x is equal to:

A 1.0

✓ Correct Answer

B 1.2**C** 0.8**D** 1.5

Explanation: $x = (a * d) / b$.

Question 39

[1 Mark]

The sum of three numbers is 105. If the ratio of the first to the second is 2 : 3 and that of the second to the third is 4 : 5, then the second number is:

A 36

✓ Correct Answer

B 24**C** 45**D** 48

Explanation: First:Second = 2:3 = 8:12. Second:Third = 4:5 = 12:15. Combined ratio = 8 : 12 : 15. Total parts = 35. Second number = $(12/35) * 105 = 36$.

Question 40

[1 Mark]

If Rs. 940 is divided into three parts proportional to $1/2 : 1/3 : 1/4$, then the first part is:

A Rs. 433.85

✓ Correct Answer

B Rs. 420**C** Rs. 450**D** Rs. 400

Explanation: Common denominator of 2, 3, 4 is 12 => ratio is 6 : 4 : 3. Sum = 13 parts. First part = $(6/13) * \text{Total}$.

Question 41

[1 Mark]

The average weight of A, B, and C is 50 kg. If the average weight of A and B is 45 kg and that of B and C is 48 kg, then the weight of B is:

A 36 kg

✓ Correct Answer

B 38 kg**C** 40 kg**D** 42 kg

Explanation: Weight of B = $(A+B) + (B+C) - (A+B+C) = 2*(45) + 2*(48) - 3*(50) = 90 + 96 - 150 = 36$ kg.

Question 42

[1 Mark]

The average weight of 40 students is 52 kg. It was found later that one weight of 35 kg was misread as 75 kg. What is the correct average?

A 51 kg

✓ Correct Answer

B 50.5 kg**C** 51.5 kg**D** 50 kg

Explanation: Sum error = Misread - Correct = 40 kg excess. Average adjustment = $-40 / 40 = -1$ kg.

Question 43

[1 Mark]

The average age of 25 students is 12 years. If the teacher's age is included, the average increases by 1 year. What is the age of the teacher?

A 38 years

✓ Correct Answer

B 36 years**C** 40 years**D** 42 years

Explanation: Teacher's age = New Average + (Initial Students * Increase in Average).

Question 44

[1 Mark]

The average marks of 150 candidates was 40. If the average of passed candidates was 45 and failed candidates was 20, the number of candidates who passed is:

A 120

✓ Correct Answer

B 110**C** 125**D** 130

Explanation: Using alligation: (Avg - Failed) : (Passed - Avg) gives the ratio of passed to failed students.

Question 45

[1 Mark]

Five years ago, the ages of A and B were in the ratio 2 : 3. At present, the ratio is 3 : 4. Find the present age of A:

A 15 years

✓ Correct Answer

B 18 years**C** 20 years**D** 24 years

Explanation: Let ages 5 yrs ago be $2x$ and $3x$. $(2x + 5)/(3x + 5) = 3/4 \Rightarrow 8x + 20 = 9x + 15 \Rightarrow x = 5$. Present age of A = $2(5) + 5 = 15$ years.

Question 46

[1 Mark]

Two years ago, the ratio of Ron and Sam's ages was 3 : 4. Six years hence, this ratio will become 5 : 6. How old is Sam now?

A 18 years

✓ Correct Answer

B 16 years**C** 20 years**D** 22 years

Explanation: Difference in years = 8. $(3x + 8)/(4x + 8) = 5/6 \Rightarrow 18x + 48 = 20x + 40 \Rightarrow 2x = 8 \Rightarrow x = 4$. Sam's present age = $4(4) + 2 = 18$ years.

Question 47

[1 Mark]

Present ages of Rahul and Amit are in the ratio 6 : 5. Four years hence, their ages will be in the ratio 7 : 6. What is Amit's present age?

A 20 years

✓ Correct Answer

B 24 years**C** 28 years**D** 32 years

Explanation: Ratio increments by 1 unit in 4 years $\Rightarrow$ 1 unit = 4 years. Amit's present age = $5 * 4 = 20$ years.

Question 48

[1 Mark]

Four years ago, a father was 6 times as old as his son. At present, the father is 4 times as old as his son. What are their present ages?

A 40, 10

✓ Correct Answer

B 36, 9**C** 48, 12**D** 32, 8

Explanation: $(4x - 4) = 6(x - 4) \Rightarrow 4x - 4 = 6x - 24 \Rightarrow 2x = 20 \Rightarrow x = 10$ (son) and 40 (father).

Question 49

[1 Mark]

Eight years ago, P was one-third of Q in age. If the ratio of their present ages is 3 : 5, what is the sum of their present ages?

A 32 years

✓ Correct Answer

B 40 years**C** 48 years**D** 56 years

Explanation: Set up equation $(3x - 8) = (1/3)(5x - 8) \Rightarrow 9x - 24 = 5x - 8 \Rightarrow 4x = 16 \Rightarrow x = 4$. Total = $8x = 32$ years.

Question 50

[1 Mark]

Ten years ago, P was half of Q in age. If the ratio of their present ages is 3 : 4, what is the total of their present ages?

A 35 years

✓ Correct Answer

B 40 years**C** 45 years**D** 50 years

Explanation: $(3x - 10) = 0.5 * (4x - 10) \Rightarrow 3x - 10 = 2x - 5 \Rightarrow x = 5$. Total = $7x = 35$ years.

Question 51

[1 Mark]

A bank offers 6% compound interest calculated on a half-yearly basis. A customer deposits Rs. 2,000 each on 1st January and 1st July of a year. At the end of the year, the amount gained by way of interest is:

A Rs. 181.80

✓ Correct Answer

B Rs. 175.00**C** Rs. 192.40**D** Rs. 160.00

Explanation: Jan 1 deposit earns interest for 2 half-years ($r=3\%$), July 1 deposit earns interest for 1 half-year.

Question 52

[1 Mark]

The difference between simple and compound interests compounded annually on a sum for 2 years at 5% per annum is Rs. 15. The sum is:

A Rs. 6,000

✓ Correct Answer

B Rs. 5,000**C** Rs. 7,200**D** Rs. 8,000

Explanation: $\text{Diff} = P * (r/100)^2 \Rightarrow P = \text{Diff} / (r/100)^2$.

Question 53

[1 Mark]

A father is aged three times more than his son Ronit. After 8 years, he would be two and a half times of Ronit's age. After further 8 years, how many times would he be of Ronit's age?

A 2 times

✓ Correct Answer

B 2.25 times**C** 2.5 times**D** 3 times

Explanation: Father = $4x$, Son = x . After 8 yrs, $4x+8 = 2.5(x+8) \Rightarrow 1.5x = 12 \Rightarrow x = 8$. Father = 32, son = 8. After 16 yrs: Father 48, son 24 $\Rightarrow$ exactly 2 times.

Question 54

[1 Mark]

A father said to his son, 'I was as old as you are at the present at the time of your birth'. If the father's age is 42 years now, the son's age five years back was:

A 16 years

✓ Correct Answer

B 18 years**C** 21 years**D** 26 years

Explanation: Let son's present age be x . Father was x at son's birth, so father's present age = $2x = 42 \Rightarrow x = 21$. Five years back, son was $21 - 5 = 16$ years.

Question 55

[1 Mark]

A card is drawn from a well-shuffled pack of 52 playing cards. What is the probability of getting a queen of clubs or a king of hearts?

A $1/26$

✓ Correct Answer

B $1/52$ **C** $2/13$ **D** $1/13$

Explanation: There is 1 queen of clubs and 1 king of hearts. Total favorable outcomes = 2. Probability = $2/52 = 1/26$.

Question 56

[1 Mark]

From a group of 6 men and 5 women, a committee of 4 persons is to be formed so that at least 2 men are on the committee. In how many ways can this be done?

A 265 ways

✓ Correct Answer

B 280 ways**C** 295 ways**D** 310 ways

Explanation: Ways = (2 Men & 2 Women) + (3 Men & 1 Woman) + (4 Men) = $({}^6C_2 * {}^5C_2) + ({}^6C_3 * {}^5C_1) + ({}^6C_4) = (15 * 10) + (20 * 5) + 15 = 150 + 100 + 15 = 265$ ways.

Question 57

[1 Mark]

In how many different ways can the letters of the word 'DETAIL' be arranged in such a way that the vowels always come together?

A 144

✓ Correct Answer

B 120**C** 240**D** 720

Explanation: Bundle the vowels into a single unit, arrange the units, and multiply by permutations of the vowels inside the bundle.

Question 58

[1 Mark]

Seeta, Reeta and Geeta spend 12%, 14% and 16% of their monthly salary on travelling respectively and each saves half of the remaining amount. Seeta and Geeta have identical monthly salaries. Seeta's monthly saving is Rs. 360 more than Geeta's, and the travelling expenditure of Seeta and Reeta together is Rs. 1240 more than that of Geeta. The monthly salary of Reeta is how much less than that of Seeta?

A Rs. 4,000 less

✓ Correct Answer

B Rs. 3,000 less**C** Rs. 2,000 less**D** Rs. 5,000 less

Explanation: Seeta/Geeta salary S : $(1/2) \cdot 0.88 S - (1/2) \cdot 0.84 S = 0.02 S = 360 \Rightarrow S = \text{Rs. } 18,000$. Travel: $0.12(18000) + 0.14(R) = 0.16(18000) + 1240 \Rightarrow 2160 + 0.14 R = 2880 + 1240 = 4120 \Rightarrow 0.14 R = 1960 \Rightarrow R = \text{Rs. } 14,000$. Reeta is Rs. 4,000 less than Seeta.

Question 59

[1 Mark]

Based on the salary data of Seeta, Reeta and Geeta (Salaries: Seeta = Rs. 18,000, Reeta = Rs. 14,000, Geeta = Rs. 18,000; travel spends 12%, 14%, 16%; savings = half of remainder), what is the sum of the savings of all three friends together?

A Rs. 21,500

✓ Correct Answer

B Rs. 22,000**C** Rs. 23,500**D** Rs. 24,000

Explanation: Savings: Seeta = $0.44 \cdot 18000 = 7920$; Reeta = $0.43 \cdot 14000 = 6020$; Geeta = $0.42 \cdot 18000 = 7560$. Total = $7920 + 6020 + 7560 = \text{Rs. } 21,500$.

Question 60

[1 Mark]

Based on the salary data of Seeta, Reeta and Geeta (Total monthly savings = Rs. 21,500, Total monthly salary = Rs. 50,000), the total monthly savings of the three friends together is what percentage of their total monthly salary?

A 43%

✓ Correct Answer

B 45%**C** 48%**D** 50%

Explanation: $(21,500 / 50,000) \cdot 100 = 43\%$.

Question 61

[1 Mark]

Rajeev's age after 15 years will be 5 times his age 5 years back. What is the present age of Rajeev?

A 10 years

✓ Correct Answer

B 8 years**C** 12 years**D** 15 years

Explanation: $R + 15 = 5(R - 5) \Rightarrow R + 15 = 5R - 25 \Rightarrow 4R = 40 \Rightarrow R = 10$ years.

Question 62

[1 Mark]

The ratio between Rahul and Deepak's ages is 4 : 3. After 6 years, Rahul's age will be 26 years. What is the present age of Deepak?

A 15 years

✓ Correct Answer

B 12 years**C** 18 years**D** 20 years

Explanation: Rahul present age = $26 - 6 = 20$ years. $4x = 20 \Rightarrow x = 5$. Deepak's present age = $3 * 5 = 15$ years.

Question 63

[1 Mark]

The ratio of a father's age to his son's age is 4 : 1. The product of their ages is 196. What will be the ratio of their ages after 5 years?

A 11 : 4

✓ Correct Answer

B 5 : 2**C** 11 : 6**D** 13 : 5

Explanation: $4x * x = 196 \Rightarrow 4x^2 = 196 \Rightarrow x^2 = 49 \Rightarrow x = 7$. Present ages: 28 and 7. After 5 years: 33 and 12 $\Rightarrow 33:12 = 11:4$.

Question 64

[1 Mark]

A is two years older than B, who is twice as old as C. If the total of the ages of A, B, and C is 27, how old is B?

A 10 years

✓ Correct Answer

B 8 years**C** 12 years**D** 14 years

Explanation: Let $C = x$, $B = 2x$, $A = 2x + 2$. Sum: $5x + 2 = 27 \Rightarrow 5x = 25 \Rightarrow x = 5$. $B = 2(5) = 10$ years.

Question 65

[1 Mark]

The average age of Ram and Shyam is 65 years. The average age of Ram, Shyam, and John is 53 years. What is the age of John?

A 29 years

✓ Correct Answer

B 27 years**C** 31 years**D** 33 years

Explanation: $\text{Ram} + \text{Shyam} = 2 * 65 = 130$. $\text{Ram} + \text{Shyam} + \text{John} = 3 * 53 = 159$. $\text{John} = 159 - 130 = 29$ years.

Question 66

[1 Mark]

A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share of the remaining profit is Rs. 855, what is the total profit?

A Rs. 1,500

✓ Correct Answer

B Rs. 1,425**C** Rs. 1,600**D** Rs. 1,750

Explanation: A gets $(3/5)$ of 95% of Profit = $0.57 * \text{Profit} = 855 \Rightarrow \text{Profit} = 855 / 0.57 = \text{Rs. } 1,500$.

Question 67

[1 Mark]

A starts a business with Rs. 3,500 and after 5 months, B joins as a partner. After a full year, the profit is divided in the ratio 2 : 3. What was B's capital contribution?

A Rs. 9,000

✓ Correct Answer

B Rs. 8,500**C** Rs. 7,500**D** Rs. 8,000

Explanation: $(3500 * 12) / (B * 7) = 2 / 3 \Rightarrow 42000 / 7B = 2/3 \Rightarrow 6000 / B = 2/3 \Rightarrow B = \text{Rs. } 9,000$.

Question 68

[1 Mark]

Simran started a business investing Rs. 50,000. After six months, Nanda joined with Rs. 80,000. After 3 years, they earned a total profit of Rs. 24,500. What was Simran's share in the profit?

A Rs. 10,500

✓ Correct Answer

B Rs. 9,423**C** Rs. 12,500**D** Rs. 14,000

Explanation: Simran: $50,000 * 36 = 1,800,000$. Nanda: $80,000 * 30 = 2,400,000$. Ratio = $18 : 24 = 3 : 4$. Simran's share = $(3/7) * 24500 = \text{Rs. } 10,500$.

Question 69

[1 Mark]

If Rs. 100,000 and Rs. 150,000 are invested by A and B respectively, and the annual profit is Rs. 24,000, what is A's share?

A Rs. 9,600

✓ Correct Answer

B Rs. 8,000**C** Rs. 10,500**D** Rs. 12,000

Explanation: Investment ratio is 2 : 3. A's share = $(2/5) * \text{Total Profit}$.

Question 70

[1 Mark]

X and Y invested in a business and divided profits in the ratio 2 : 3. If X invested Rs. 40,000, find the capital invested by Y:

A Rs. 60,000

✓ Correct Answer

B Rs. 50,000**C** Rs. 75,000**D** Rs. 80,000

Explanation: $40000 / Y = 2/3 \Rightarrow Y = \text{Rs. } 60,000$.

Question 71

[1 Mark]

A, B, and C started a business by investing Rs. 25,000, Rs. 30,000, and Rs. 35,000 respectively. Find the share of B out of an annual profit of Rs. 187,200:

A Rs. 62,400

✓ Correct Answer

B Rs. 52,000**C** Rs. 72,800**D** Rs. 65,400

Explanation: Ratio = 25 : 30 : 35 = 5 : 6 : 7. Total parts = 18. B's share = $(6/18) * 187200 = (1/3) * 187200 = \text{Rs. } 62,400$.

Question 72

[1 Mark]

Rs. 1,400 is divided among Amla, Bimla, and Simla so that Amla receives half as much as Bimla, and Bimla receives half as much as Simla. What is Simla's share?

A Rs. 800

✓ Correct Answer

B Rs. 400**C** Rs. 600**D** Rs. 700

Explanation: Let Simla = $4x$, Bimla = $2x$, Amla = x . Total = $7x = 1400 \Rightarrow x = 200$. Simla = $4(200) = \text{Rs. } 800$.

Question 73

[1 Mark]

Smith and Kate started a venture investing Rs. 84,000 and Rs. 28,000 respectively. In what ratio should the profit earned after 2 years be divided between Smith and Kate?

A 3 : 1

✓ Correct Answer

B 2 : 1**C** 4 : 1**D** 5 : 2

Explanation: $84000 : 28000 = 84 : 28 = 3 : 1$.

Question 74

[1 Mark]

Sham and Ram gained profit divided in the ratio 2 : 3. If Sham invested Rs. 40,000, what was the capital invested by Ram?

A Rs. 60,000

✓ Correct Answer

B Rs. 50,000**C** Rs. 75,000**D** Rs. 80,000

Explanation: $40000 / \text{Ram} = 2/3 \Rightarrow \text{Ram} = \text{Rs. } 60,000$.

Question 75

[1 Mark]

Harry, John, and Smith start a business investing Rs. 27,000, Rs. 72,000, and Rs. 81,000 respectively. At the end of the year, Smith earns a profit share of Rs. 36,000. Find the total profit:

A Rs. 80,000

✓ Correct Answer

B Rs. 75,000**C** Rs. 90,000**D** Rs. 85,000

Explanation: Ratio = 27 : 72 : 81 = 3 : 8 : 9. Total parts = 20. Smith share = $(9/20) \times \text{Total} = 36000 \Rightarrow \text{Total} = 36000 \times 20 / 9 = \text{Rs. } 80,000$.

Question 76

[1 Mark]

In what ratio should two varieties of sugar of Rs. 18/kg and Rs. 24/kg be mixed together to get a mixture whose cost is Rs. 20/kg?

A 2 : 1

✓ Correct Answer

B 1 : 2**C** 3 : 1**D** 4 : 1

Explanation: Using alligation: $(24 - 20) : (20 - 18) = 4 : 2 = 2 : 1$.

Question 77

[1 Mark]

Two vessels A and B contain spirit and water in the ratio 5 : 2 and 7 : 6 respectively. Find the ratio in which these mixtures must be combined to obtain a new mixture containing spirit and water in the ratio 8 : 5:

A 7 : 9

✓ Correct Answer

B 5 : 7**C** 6 : 7**D** 8 : 9

Explanation: Fraction of spirit: A = $5/7$, B = $7/13$, Mean = $8/13$. Alligation gives $(8/13 - 7/13) : (5/7 - 8/13) = 1/13 : 9/91 = 7 : 9$.

Question 78

[1 Mark]

How much water must be added to 40 litres of milk at the cost price of Rs. 3.50 per litre so that the cost of the diluted milk is reduced to Rs. 2.00 per litre?

A 30 litres

✓ Correct Answer

B 25 litres**C** 35 litres**D** 20 litres

Explanation: Total cost = $40 \times 3.50 = 140$. New volume = $140 / 2 = 70$ litres. Water added = $70 - 40 = 30$ litres.

Question 79

[1 Mark]

A dishonest milkman professes to sell milk at cost price but mixes it with water and thereby gains 25%. What is the percentage of water in the final mixture?

A 20%

✓ Correct Answer

B 25%**C** 15%**D** 30%

Explanation: Water : Milk = $25 : 100 = 1 : 4$. Percentage of water in mixture = $1 / (1 + 4) \times 100 = 20\%$.

Question 80

[1 Mark]

In a 729-litre mixture of milk and water, the ratio of milk to water is 7 : 2. To obtain a new mixture containing milk and water in the ratio 7 : 3, what amount of water must be added?

A 81 litres

✓ Correct Answer

B 75 litres**C** 90 litres**D** 65 litres

Explanation: Total parts = 9. 1 part = $729 / 9 = 81$ litres. Milk is 7 parts (567L), water is 2 parts (162L). To make ratio 7:3, water needed is 3 parts = 243L. Water added = $243 - 162 = 81$ litres.

Question 81

[1 Mark]

In what ratio must wheat A at Rs. 10.50 per kg be mixed with wheat B at Rs. 12.30 per kg so that the mixture is worth Rs. 11.00 per kg?

A 13 : 5

✓ Correct Answer

B 11 : 5**C** 12 : 7**D** 14 : 5

Explanation: Alligation: $(12.30 - 11.00) : (11.00 - 10.50) = 1.30 : 0.50 = 13 : 5$.

Question 82

[1 Mark]

In what ratio must a shopkeeper mix Peas at Rs. 16/kg and Soybeans at Rs. 25/kg to obtain a mixture worth Rs. 19.50/kg?

A 11 : 7

✓ Correct Answer

B 9 : 5**C** 7 : 5**D** 12 : 8

Explanation: Alligation: $(25 - 19.50) : (19.50 - 16) = 5.5 : 3.5 = 11 : 7$.

Question 83

[1 Mark]

Two containers P and Q contain milk and water in the ratio 5 : 2 and 7 : 6 respectively. Find the ratio in which these two mixtures should be combined to produce a mixture R with ratio 8 : 5:

A 7 : 9

✓ Correct Answer

B 5 : 6**C** 4 : 9**D** 9 : 7

Explanation: By alligation on milk concentration: (5/7) and (7/13) to obtain (8/13) yields 7 : 9.

Question 84

[1 Mark]

A container contains 30 litres of pure milk. From this container, 5 litres of milk is taken out and replaced by water. This process is repeated further two times. How much pure milk remains in the container?

A 17.36 litres

✓ Correct Answer

B 18.45 litres**C** 21.87 litres**D** 23.44 litres

Explanation: Remaining milk = $30 * (1 - 5/30)^3 = 30 * (5/6)^3 = 30 * (125/216) \approx 17.36$ litres.

Question 85

[1 Mark]

What will be the simple interest on Rs. 80,000 at 10% per annum for 9 months?

A Rs. 6,000

✓ Correct Answer

B Rs. 8,000**C** Rs. 7,500**D** Rs. 5,400

Explanation: SI = $(P * R * T) / 100$ where $T = 9/12 = 0.75$ years.

Question 86

[1 Mark]

At what rate percent per annum will a sum of money double itself in 20 years at simple interest?

A 5%

✓ Correct Answer

B 4%**C** 6%**D** 8%

Explanation: To double, Interest = P $\Rightarrow$ $P = (P * R * T)/100 \Rightarrow R = 100 / T$.

Question 87

[1 Mark]

A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The principal sum is:

A Rs. 698

✓ Correct Answer

B Rs. 650**C** Rs. 690**D** Rs. 700

Explanation: 1 year interest = $854 - 815 = \text{Rs. } 39$. 3 years interest = $3 * 39 = \text{Rs. } 117$. Principal = $815 - 117 = \text{Rs. } 698$.

Question 88

[1 Mark]

A sum of Rs. 12,500 amounts to Rs. 15,500 in 4 years at simple interest. What is the annual rate of interest?

A 6%

✓ Correct Answer

B 4%**C** 5%**D** 7%

Explanation: Interest = $15500 - 12500 = 3000$. Rate = $(3000 * 100) / (12500 * 4) = 300000 / 50000 = 6\%$.

Question 89

[1 Mark]

A, B, and C invest Rs. 20,000, Rs. 25,000, and Rs. 35,000 in a business. Find the share of B out of a total profit of Rs. 128,000:

A Rs. 40,000

✓ Correct Answer

B Rs. 32,000**C** Rs. 45,000**D** Rs. 48,000

Explanation: Ratio = 4 : 5 : 7. Total parts = 16. B's share = $(5/16) * 128000 = \text{Rs. } 40,000$.

Question 90

[1 Mark]

Rs. 2,100 is divided among X, Y, and Z such that X receives half as much as Y, and Y receives half as much as Z. What is Z's share?

A Rs. 1,200

✓ Correct Answer

B Rs. 600**C** Rs. 900**D** Rs. 1,000

Explanation: $X:Y:Z = 1:2:4$. Total parts = 7. Z's share = $(4/7) * 2100 = \text{Rs. } 1,200$.

Question 91

[1 Mark]

P and Q start a business with Rs. 45,000 and Rs. 15,000 respectively. In what ratio should the profit after 1 year be divided?

A 3 : 1

✓ Correct Answer

B 2 : 1**C** 4 : 1**D** 5 : 2

Explanation: $45000 : 15000 = 3 : 1$.

Question 92

[1 Mark]

A and B divide profit in the ratio 3 : 5. If A invested Rs. 45,000, what was B's investment?

A Rs. 75,000

✓ Correct Answer

B Rs. 60,000**C** Rs. 80,000**D** Rs. 90,000

Explanation: $45000 / B = 3/5 \Rightarrow B = \text{Rs. } 75,000$.

Question 93

[1 Mark]

Three partners invest in the ratio 2 : 5 : 8. The third partner earns Rs. 48,000 as profit. What is the total profit?

A Rs. 90,000

✓ Correct Answer

B Rs. 80,000**C** Rs. 96,000**D** Rs. 100,000

Explanation: 8 parts = 48000 $\Rightarrow$ 1 part = 6000. Total = 15 parts = Rs. 90,000.

Question 94

[1 Mark]

A shopkeeper sells an item for Rs. 240 at a loss of 20%. What is the cost price of the article?

A Rs. 300

✓ Correct Answer

B Rs. 280**C** Rs. 320**D** Rs. 350

Explanation: $SP = 0.80 * CP \Rightarrow CP = 240 / 0.80 = \text{Rs. } 300.$

Question 95

[1 Mark]

The selling price of 10 candles is equal to the cost price of 12 candles. What is the gain percent?

A 20%

✓ Correct Answer

B 15%**C** 25%**D** 18%

Explanation: $\text{Gain \%} = ((12 - 10) / 10) * 100 = 20\%.$

Question 96

[1 Mark]

The selling price of 40 articles is equal to the cost price of 35 articles. Find the profit or loss percent:

A 12.5% loss

✓ Correct Answer

B 12.5% gain**C** 14.2% loss**D** 10% loss

Explanation: $\text{Loss \%} = ((40 - 35) / 40) * 100 = (5/40) * 100 = 12.5\% \text{ loss.}$

Question 97

[1 Mark]

If a retailer marks up goods by 25% and then offers a 20% discount on the marked price, what is the net profit or loss percentage?

A 0% (No profit no loss)

✓ Correct Answer

B 5% gain**C** 5% loss**D** 2% gain

Explanation: $\text{Factor} = 1.25 * 0.80 = 1.00 \Rightarrow \text{Exactly } 0\% \text{ profit or loss.}$

Question 98

[1 Mark]

A merchant made a profit of 25% by selling an outfit for Rs. 5,000. What was the cost price of the outfit?

A Rs. 4,000

✓ Correct Answer

B Rs. 3,750**C** Rs. 4,200**D** Rs. 4,500

Explanation: $CP = 5000 / 1.25 = \text{Rs. } 4,000.$

Question 99

[1 Mark]

In what ratio must rice at Rs. 24/kg be mixed with premium rice at Rs. 34/kg to obtain a blend worth Rs. 30/kg?

A 2 : 3

✓ Correct Answer

B 3 : 2**C** 1 : 2**D** 3 : 4

Explanation: Alligation: $(34 - 30) : (30 - 24) = 4 : 6 = 2 : 3$.

Question 100

[1 Mark]

In what ratio should cashews at Rs. 400/kg be mixed with almonds at Rs. 600/kg to achieve a blend priced at Rs. 480/kg?

A 3 : 2

✓ Correct Answer

B 2 : 3**C** 1 : 1**D** 4 : 3

Explanation: Alligation: $(600 - 480) : (480 - 400) = 120 : 80 = 3 : 2$.

Question 101

[1 Mark]

Two containers have acid and water in ratios 3 : 1 and 5 : 3. In what ratio must they be mixed to obtain a 2 : 1 ratio?

A 1 : 1

✓ Correct Answer

B 2 : 1**C** 3 : 2**D** 1 : 2

Explanation: Using alligation on acid proportions gives 1 : 1.

Question 102

[1 Mark]

A vat contains 40 litres of chemical solution. 4 litres are drawn off and replaced with water. This operation is repeated twice more. How much of the original chemical remains?

A 29.16 litres

✓ Correct Answer

B 28.00 litres**C** 30.24 litres**D** 27.50 litres

Explanation: Remaining = $40 * (1 - 4/40)^3 = 40 * (0.9)^3 = 40 * 0.729 = 29.16$ litres.

Question 103

[1 Mark]

Find the simple interest on Rs. 45,000 at 8% per annum for 6 months:

A Rs. 1,800

✓ Correct Answer

B Rs. 2,400**C** Rs. 1,600**D** Rs. 2,100

Explanation: $SI = 45000 * 0.08 * 0.5 = \text{Rs. } 1,800.$

Question 104

[1 Mark]

At what simple interest rate per annum will an initial deposit double in 10 years?

A 10%

✓ Correct Answer

B 8%**C** 12%**D** 15%

Explanation: $\text{Rate} = 100 / 10 = 10\%.$

Question 105

[1 Mark]

A sum at simple interest amounts to Rs. 920 in 2 years and Rs. 1,040 in 3 years. What is the initial principal?

A Rs. 680

✓ Correct Answer

B Rs. 720**C** Rs. 640**D** Rs. 700

Explanation: $1 \text{ year interest} = 1040 - 920 = 120. \text{Principal} = 920 - 2(120) = \text{Rs. } 680.$

Question 106

[1 Mark]

A principal of Rs. 8,000 yields Rs. 1,920 interest over 3 years. What is the rate of interest per annum?

A 8%

✓ Correct Answer

B 6%**C** 7.5%**D** 9%

Explanation: $\text{Rate} = (1920 * 100) / (8000 * 3) = 192000 / 24000 = 8\%.$

Question 107

[1 Mark]

Find the missing term in the sequence: ABC, ABC2, AB2C2, ____, A2B2C3

A A2B2C2

✓ Correct Answer

B ABC3**C** AB3C2**D** A3B2C2

Explanation: Digits increment sequentially on each letter from left to right.

Question 108

[1 Mark]

Find the next term in the series: BLL, DNN, FPP, ____, JTT

A HRR

✓ Correct Answer

B GQQ**C** HSS**D** IRR

Explanation: First letter advances by +2 (B->D->F->H->J), second and third advance by +2 (L->N->P->R->T).

Question 109

[1 Mark]

Look at the series: 3, 9, 27, 81, ... What number should come next?

A 243

✓ Correct Answer

B 162**C** 216**D** 324

Explanation: Each term is multiplied by 3 ($81 * 3 = 243$).

Question 110

[1 Mark]

Look at the sequence: 5, 4, 6, 15, 56, ... What number should come next?

A 275

✓ Correct Answer

B 265**C** 280**D** 290

Explanation: Pattern: $5*1 - 1 = 4$; $4*2 - 2 = 6$; $6*3 - 3 = 15$; $15*4 - 4 = 56$; $56*5 - 5 = 275$.

Question 111

[1 Mark]

Statements: All desks are chairs. No chair is a table.

Conclusions:

(I) Some desks are tables.

(II) No table is a desk.

A Only (II) follows

✓ Correct Answer

B Only (I) follows**C** Both follow**D** Neither follows

Explanation: Since all desks are inside chairs and disjoint from tables, no desk can be a table. Hence, conclusion (II) follows.

Question 112

[1 Mark]

Given the rules:

A # B means B is 1m East of A

A \$ B means B is 1m North of A

A * B means B is 1m West of A

A @ B means B is 1m South of A

According to $P @ Q * R$, in which direction is R with respect to P?

A South-West

✓ Correct Answer

B North-East**C** South-East**D** North-West

Explanation: Q is 1m South of P. R is 1m West of Q. Hence, R is South-West of P.

Question 113

[1 Mark]

Using the same rules (# = East, \$ = North, * = West, @ = South):

According to $J \# K \$ L$, in which direction is L with respect to J?

A North-East

✓ Correct Answer

B North-West**C** South-East**D** South-West

Explanation: K is 1m East of J. L is 1m North of K. L is North-East of J.

Question 114

[1 Mark]

Look at this series: 34, 33, 35, 34, 36, 35, ... What number should come next?

A 37

✓ Correct Answer

B 36**C** 38**D** 34

Explanation: Alternating pattern: -1, +2, -1, +2, -1, +2... Next is $35 + 2 = 37$.

Question 115

[1 Mark]

Look at this series: 100, 20, 90, 25, 80, ... What number should come next?

A 30

✓ Correct Answer

B 35**C** 70**D** 75

Explanation: Odd terms decrease by 10 (100, 90, 80). Even terms increase by 5 (20, 25, 30).

Question 116

[1 Mark]

One morning after sunrise, Aarav and Varun were talking face to face. If Varun's shadow fell exactly to the left of Aarav, which direction was Aarav facing?

A North

✓ Correct Answer

B South**C** East**D** West

Explanation: In the morning, the sun is in the East, so shadows fall towards the West. If the shadow is to Aarav's left, Aarav is facing North.

Question 117

[1 Mark]

If South-East becomes North, North-East becomes West, and so on. What will West become in this coordinate system?

A South-East

✓ Correct Answer

B North-West**C** South-West**D** North-East

Explanation: The compass rotates 135 degrees clockwise. West rotated 135 degrees clockwise becomes South-East.

Question 118

[1 Mark]

A clock is placed such that at 6:00 PM, the hour hand points North. In which direction will the minute hand point at 9:15 PM?

A West

✓ Correct Answer

B East**C** North**D** South

Explanation: At 6:00, the 6-position points North (rotated 180 deg). At 9:15, the minute hand is at 3, which is opposite to 9 (pointing West).

Question 119

[1 Mark]

From home, Maya walked 20 km North, turned West and walked 15 km, turned South and walked 8 km, then turned East and walked 15 km. In which direction and distance is she from her home?

A 12 km North

✓ Correct Answer

B 12 km South**C** 15 km North**D** 8 km East

Explanation: East/West displacements cancel (15 km W, 15 km E). Net North displacement = $20 - 8 = 12$ km North.

Question 120

[1 Mark]

Pointing to a photograph of a boy, Ramesh said, 'He is the son of the only son of my mother.' How is Ramesh related to the boy?

A Father

✓ Correct Answer

B Uncle**C** Brother**D** Grandfather

Explanation: Only son of Ramesh's mother is Ramesh himself. The boy is the son of Ramesh. Ramesh is his Father.

Question 121

[1 Mark]

Pointing to a photograph, Meera said, 'He is the son of the only son of my paternal grandfather.' How is the man in the photograph related to Meera?

A Brother

✓ Correct Answer

B Father**C** Cousin**D** Uncle

Explanation: The only son of Meera's paternal grandfather is Meera's father. The son of Meera's father is Meera's Brother.

Question 122

[1 Mark]

Consider the triangular pattern where bottom numbers (A, B, C) relate to the apex number:

Row 1: (5, 4, 7) -> 8

Row 2: (6, 9, 5) -> 10

Row 3: (3, 7, 2) -> ?

What number replaces the question mark?

A 6

✓ Correct Answer

B 8**C** 10**D** 12

Explanation: Pattern: $(A * C) / B$ or $(\text{Sum of outer} / 2) = 6$.

Question 123

[1 Mark]

Statements: All pens are papers. All papers are folders.

Conclusions:

(I) Some folders are pens.

(II) All pens are folders.

A Both follow

✓ Correct Answer

B Only (I) follows**C** Only (II) follows**D** Neither follows

Explanation: Both conclusions follow from transitive inclusion.

Question 124

[1 Mark]

Statements: All birds are fliers. Some kites are fliers.

Conclusions:

(I) All birds are kites.

(II) Some kites are birds.

A Neither follows

✓ Correct Answer

B Only (I) follows

C Only (II) follows

D Both follow

Explanation: No definite link exists between birds and kites.

Question 125

[1 Mark]

Look at the sequence: 589654237, 89654237, 8965423, 965423, ?

What comes next?

A 96542

✓ Correct Answer

B 96543

C 65423

D 96785

Explanation: The pattern alternates dropping the first digit, then dropping the last digit. Next is dropping the last digit 3 from 965423 => 96542.

Question 126

[1 Mark]

Look at the sequence: 11, 10, ?, 100, 1001, 1000, 10001

What term replaces the question mark?

A 101

✓ Correct Answer

B 110

C 111

D 1001

Explanation: The sequence alternates between $(10^k + 1)$ and 10^k . 11, 10, 101, 100, 1001, 1000, 10001. Missing is 101.

Question 127

[1 Mark]

Data Interpretation: A family's monthly budget is Rs. 60,000 allocated as: Food (20%), Savings (15%), Clothing (15%), Housing (10%), Education (10%), Transport (15%), Miscellaneous (15%). How much is spent on Food?

A Rs. 12,000

✓ Correct Answer

B Rs. 10,000

C Rs. 14,000

D Rs. 15,000

Explanation: 20% of 60,000 = Rs. 12,000.

Question 128

[1 Mark]

From the budget (Rs. 60,000 total: Clothing 15%, Housing 10%, Transport 15%), how much more money is spent on Clothing and Housing together than on Transport?

A Rs. 6,000

✓ Correct Answer

B Rs. 4,500**C** Rs. 7,500**D** Rs. 9,000

Explanation: $(15\% + 10\%) - 15\% = 10\%$. 10% of 60,000 = Rs. 6,000.

Question 129

[1 Mark]

From the budget (Food = 20%, Education = 10%), what percentage of the amount spent on food is spent on education?

A 50%

✓ Correct Answer

B 25%**C** 40%**D** 60%

Explanation: $(10\% / 20\%) * 100 = 50\%$.

Question 130

[1 Mark]

In a pie chart representing sales by segment: Segment A (30%), Segment B (25%), Segment C (35%), Segment D (10%). What is the central angle corresponding to Segment A?

A 108°

✓ Correct Answer

B 90° **C** 126° **D** 120°

Explanation: Angle = 30% of $360^\circ = 0.30 * 360^\circ = 108^\circ$.

Question 131

[1 Mark]

Solve for A, B, C in the cryptarithmic puzzle: $ABC + CBA = 444$ where distinct letters represent distinct non-zero digits. A possible solution is:

A A=1, B=2, C=3

✓ Correct Answer

B A=2, B=4, C=2**C** A=3, B=1, C=1**D** A=1, B=7, C=2

Explanation: $123 + 321 = 444$. Hence A=1, B=2, C=3.

Question 132

[1 Mark]

Solve for D, E, F in: $DEF - FED = 297$ where $D=5$, $F=2$. What is the value of E?

A Any single digit 0-9

✓ Correct Answer

B $E=4$ **C** $E=7$ **D** $E=0$

Explanation: $(500 + 10E + 2) - (200 + 10E + 5) = 502 - 205 = 297$, independent of E.

Question 133

[1 Mark]

Solve for X, Y, Z in: $XYZ + XYZ = 468$ with unique digits. Find X, Y, Z:

A $X=2, Y=3, Z=4$

✓ Correct Answer

B $X=2, Y=4, Z=2$ **C** $X=3, Y=2, Z=4$ **D** $X=1, Y=8, Z=4$

Explanation: $468 / 2 = 234 \Rightarrow X=2, Y=3, Z=4$.

Question 134

[1 Mark]

Solve for X, Y, Z in: $XYZ + XYZ = 756$ with digits. Find X, Y, Z:

A $X=3, Y=7, Z=8$

✓ Correct Answer

B $X=3, Y=6, Z=8$ **C** $X=4, Y=7, Z=8$ **D** $X=3, Y=8, Z=7$

Explanation: $756 / 2 = 378 \Rightarrow X=3, Y=7, Z=8$.

Question 135

[1 Mark]

In a survey of a town represented by a 3-circle Venn Diagram (Educated, Employed, Backward):

Educated only = 8

Educated & Employed only = 6

All three = 3

Employed only = 7

Backward only = 17

Backward & Employed only = 5

Backward & Educated only = 11

How many educated people are employed?

A 9

✓ Correct Answer

B 6**C** 14**D** 17

Explanation: Educated & Employed = (Educated & Employed only) + (All three) = $6 + 3 = 9$.

Question 136

[1 Mark]

Using the same Venn Diagram data (Educated only = 8, All three = 3, Backward & Educated only = 11), how many backward people are educated?

A 14

✓ Correct Answer

B 11**C** 9**D** 18**Explanation:** Backward & Educated = $11 + 3 = 14$.**Question 137**

[1 Mark]

Using the same Venn Diagram data, how many backward uneducated people are employed?

A 5

✓ Correct Answer

B 8**C** 11**D** 14**Explanation:** Backward & Employed (excluding Educated) = 5.**Question 138**

[1 Mark]

Using the same Venn Diagram data, how many backward people are not educated?

A 22

✓ Correct Answer

B 17**C** 20**D** 25**Explanation:** Backward not educated = (Backward only) + (Backward & Employed only) = $17 + 5 = 22$.**Question 139**

[1 Mark]

Eight persons (L, M, N, O, P, Q, R, S) sit in a circle facing inward. N is between L and R. P is second left of N. P and S have 2 persons between them. M neighbors O. One person sits between M and Q. What is O's position relative to S?

A Third to the left

✓ Correct Answer

B Immediate right**C** Second to the right**D** Immediate left**Explanation:** Arranging clockwise gives unique valid placement with O third to the left of S.

Question 140

[1 Mark]

Based on the 8-person circular table arrangement (L, M, N, O, P, Q, R, S), four pairs share an adjacent relationship. Which pair does NOT belong to the group?

A RP

✓ Correct Answer

B QL**C** NM**D** MS

Explanation: RP are not directly adjacent whereas the other pairs are adjacent.

Question 141

[1 Mark]

In the 8-person circular seating arrangement, who sits second to the left of the person immediately to the right of O?

A Q

✓ Correct Answer

B M**C** S**D** L

Explanation: Following step-by-step positions resolves to Q.

Question 142

[1 Mark]

In the 8-person circular seating arrangement, who sits directly between M and Q?

A O

✓ Correct Answer

B L**C** N**D** S

Explanation: Between M and Q sits O.

Question 143

[1 Mark]

In the 8-person circular seating arrangement, who sits third to the right of the person opposite N?

A L

✓ Correct Answer

B S**C** R**D** O

Explanation: Resolves to L.

Question 144

[1 Mark]

Seven friends (P, Q, R, S, T, U, V) sit in a circle facing the center. V is 2nd left of S and neighbors T and U. S is not a neighbor of R or T. P neighbors Q and R. Which of the following is correct?

A Q is between P and S

✓ Correct Answer

B S is between U and P**C** T is immediate right of V**D** U is immediate left of V

Explanation: Deducing the 7-seat circle confirms Q is placed between P and S.

Question 145

[1 Mark]

In the 7-friends circular seating arrangement, in which of the following pairs is the second person sitting to the immediate right of the first?

A QU

✓ Correct Answer

B VU**C** TR**D** PT

Explanation: QU matches the exact clockwise right adjacent position.

Question 146

[1 Mark]

If T and S interchange their places in the 7-friends circle, what will be S's new position?

A Neighbors of V and R

✓ Correct Answer

B Immediate left of R**C** Immediate right of U**D** Neighbors of R and P

Explanation: S will now sit adjacent to V and R.

Question 147

[1 Mark]

In the 7-friends circular seating arrangement (with friends seated clockwise as S, U, V, T, R, P, Q facing center), what is the position of R?

A Second to the right of Q

✓ Correct Answer

B Third to the left of U**C** To the immediate right of P**D** Second to the left of V

Explanation: In the clockwise circle S, U, V, T, R, P, Q facing inward: looking from Q facing center, moving counter-clockwise (to Q's right) gives Q → P → R. Thus, R is second to the right of Q (and to the immediate right of P).

Question 148

[1 Mark]

Y is to the East of X, which is to the North of Z. If P is to the South of Z, then in which direction of Y is P located?

A South-West

✓ Correct Answer

B South-East**C** North-West**D** South

Explanation: X is North of Z => Z is South of X. Y is East of X. P is South of Z => P is South and West of Y => South-West.

Question 149

[1 Mark]

A person walks 6 km South, turns right and walks 4 km, then turns left and walks 6 km. In which direction is the person from the starting point?

A South-West

✓ Correct Answer

B South-East**C** North-West**D** West

Explanation: South 6 km -> Right (West) 4 km -> Left (South) 6 km. Total: 12 km South, 4 km West => South-West.

Question 150

[1 Mark]

Rasik walked 20 m North. He turned right and walked 30 m. Then he turned right and walked 35 m. Then he turned left and walked 15 m. Finally, he turned left and walked 15 m. In which direction and how many metres is he from his starting position?

A 45 m East

✓ Correct Answer

B 30 m East**C** 45 m West**D** 30 m West

Explanation: North 20m -> East 30m -> South 35m (net 15m South) -> East 15m (net 45m East) -> North 15m (cancels South 15m). Final position = 45 m East.